Smart Hiring: An NLP and Ensemble-Based Approach to AI-Driven Resume Screening

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Abstract: AI-Powered resume screening has emerged as a vital tool to streamline recruitment processes, yet current solutions often struggle with balancing accuracy, contextual understanding, and interpretability. Addressing this gap, this research presents a hybrid methodology that combines natural language processing (NLP), deep learning, and ensemble machine learning models to enhance candidate-job matching. The system processes a structured and unstructured recruitment dataset, applying preprocessing techniques such as lemmatization, TF-IDF encoding, and contextual embeddings using BERT. Individual models including Support Vector Machine (SVM), Logistic Regression, and Named Entity Recognition (NER) are evaluated and further integrated through a stacking ensemble framework. Experimental results show the ensemble model significantly outperforms individual classifiers, achieving 94% accuracy, 92% recall, and an F1-score of 92.5%. These findings demonstrate that combining lexical, semantic, and statistical perspectives leads to improved performance and reduced classification error. The proposed system offers a scalable and intelligent solution for automated resume screening, contributing meaningfully to the fields of human resource analytics and AI-assisted recruitment.

1 INTRODUCTION

In today's fast-paced and digitally transformed hiring landscape, organizations face the challenge of efficiently filtering through thousands of resumes to identify the most suitable candidates for a given role. According to recent industry surveys, recruiters spend an average of just 6–8 seconds reviewing each

resource management. Numerous studies have explored the application of machine learning algorithms for resume classification, job matching, and candidate ranking. Early approaches primarily relied on keyword- based matching and TF-IDF representations, which, while effective to an extent, struggled to capture contextual and semantic nuances. More recent advances incorporate deep learning